



TECHNICAL LEARNING FILE

DEEP-06

Probability and Statistics

Uncertainty, estimation, and calibration

FORMAT	LENGTH	ACCESS
INTENSIVE FILE	50 PAGES	OFFLINE

Edition 2 - original curriculum - editorial review recommended



FILE / ORIENTATION

Purpose

01 FILE GUIDANCE

This optional advanced guide does not gate the core learning path. Apply probability and statistics with explicit assumptions, tests, tradeoffs, and limitations.

02 LOCAL USE

Index this page in the offline database and preserve its resource and page identifiers for bookmarks, notes, highlights, and progress.



FILE / ORIENTATION

Prerequisites

01 FILE GUIDANCE

Complete the related core track or pass its diagnostic.

NOUS AI ACADEMY

02 LOCAL USE

Index this page in the offline database and preserve its resource and page identifiers for bookmarks, notes, highlights, and progress.



FILE / ORIENTATION

Study Method

01 FILE GUIDANCE

For each unit, explain the concept, follow the workflow, reproduce the example, analyze a failure, and complete the applied lab. Record evidence locally.

02 LOCAL USE

Index this page in the offline database and preserve its resource and page identifiers for bookmarks, notes, highlights, and progress.



UNIT 01 / CONCEPT

Counting and Probability: Concept

01 OBJECTIVE

Explain and apply counting and probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Counting and Probability is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Counting, Probability are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should counting and probability be used, and what observable evidence would make that choice defensible? Build a small local example that applies counting and probability, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Counting and Probability in one precise sentence and name one assumption.
2. Create a two-step example of counting and probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 01 / WORKFLOW

Counting and Probability: Workflow

01 OBJECTIVE

Explain and apply counting and probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Counting and Probability is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to counting and probability.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies counting and probability, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Counting and Probability in one precise sentence and name one assumption.
2. Create a two-step example of counting and probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 01 / WORKED EXAMPLE

Counting and Probability: Worked Example

01 OBJECTIVE

Explain and apply counting and probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies counting and probability, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns counting and probability into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Counting and Probability in one precise sentence and name one assumption.
2. Create a two-step example of counting and probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 01 / FAILURE ANALYSIS

Counting and Probability: Failure Analysis

01 OBJECTIVE

Explain and apply counting and probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how counting and probability can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to counting and probability. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Counting and Probability in one precise sentence and name one assumption.
2. Create a two-step example of counting and probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 01 / APPLIED LAB

Counting and Probability: Applied Lab

01 OBJECTIVE

Explain and apply counting and probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of counting and probability into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies counting and probability, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Counting and Probability in one precise sentence and name one assumption.
2. Create a two-step example of counting and probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 02 / CONCEPT

Conditional Probability: Concept

01 OBJECTIVE

Explain and apply conditional probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Conditional Probability is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Conditional, Probability are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should conditional probability be used, and what observable evidence would make that choice defensible? Build a small local example that applies conditional probability, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Conditional Probability in one precise sentence and name one assumption.
2. Create a two-step example of conditional probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 02 / WORKFLOW

Conditional Probability: Workflow

01 OBJECTIVE

Explain and apply conditional probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Conditional Probability is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to conditional probability.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies conditional probability, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Conditional Probability in one precise sentence and name one assumption.
2. Create a two-step example of conditional probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 02 / WORKED EXAMPLE

Conditional Probability: Worked Example

01 OBJECTIVE

Explain and apply conditional probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies conditional probability, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns conditional probability into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Conditional Probability in one precise sentence and name one assumption.
2. Create a two-step example of conditional probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 02 / FAILURE ANALYSIS

Conditional Probability: Failure Analysis

01 OBJECTIVE

Explain and apply conditional probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how conditional probability can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to conditional probability. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Conditional Probability in one precise sentence and name one assumption.
2. Create a two-step example of conditional probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 02 / APPLIED LAB

Conditional Probability: Applied Lab

01 OBJECTIVE

Explain and apply conditional probability using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of conditional probability into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies conditional probability, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Conditional Probability in one precise sentence and name one assumption.
2. Create a two-step example of conditional probability and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 03 / CONCEPT

Random Variables: Concept

01 OBJECTIVE

Explain and apply random variables using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Random Variables is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Random, Variables are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should random variables be used, and what observable evidence would make that choice defensible? Build a small local example that applies random variables, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Random Variables in one precise sentence and name one assumption.
2. Create a two-step example of random variables and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 03 / WORKFLOW

Random Variables: Workflow

01 OBJECTIVE

Explain and apply random variables using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Random Variables is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to random variables.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies random variables, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Random Variables in one precise sentence and name one assumption.
2. Create a two-step example of random variables and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 03 / WORKED EXAMPLE

Random Variables: Worked Example

01 OBJECTIVE

Explain and apply random variables using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies random variables, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns random variables into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Random Variables in one precise sentence and name one assumption.
2. Create a two-step example of random variables and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 03 / FAILURE ANALYSIS

Random Variables: Failure Analysis

01 OBJECTIVE

Explain and apply random variables using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how random variables can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to random variables. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Random Variables in one precise sentence and name one assumption.
2. Create a two-step example of random variables and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 03 / APPLIED LAB

Random Variables: Applied Lab

01 OBJECTIVE

Explain and apply random variables using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of random variables into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies random variables, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Random Variables in one precise sentence and name one assumption.
2. Create a two-step example of random variables and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 04 / CONCEPT

Common Distributions: Concept

01 OBJECTIVE

Explain and apply common distributions using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Common Distributions is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Common, Distributions are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should common distributions be used, and what observable evidence would make that choice defensible? Build a small local example that applies common distributions, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Common Distributions in one precise sentence and name one assumption.
2. Create a two-step example of common distributions and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 04 / WORKFLOW

Common Distributions: Workflow

01 OBJECTIVE

Explain and apply common distributions using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Common Distributions is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to common distributions.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies common distributions, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Common Distributions in one precise sentence and name one assumption.
2. Create a two-step example of common distributions and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 04 / WORKED EXAMPLE

Common Distributions: Worked Example

01 OBJECTIVE

Explain and apply common distributions using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies common distributions, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns common distributions into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Common Distributions in one precise sentence and name one assumption.
2. Create a two-step example of common distributions and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 04 / FAILURE ANALYSIS

Common Distributions: Failure Analysis

01 OBJECTIVE

Explain and apply common distributions using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how common distributions can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to common distributions. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Common Distributions in one precise sentence and name one assumption.
2. Create a two-step example of common distributions and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 04 / APPLIED LAB

Common Distributions: Applied Lab

01 OBJECTIVE

Explain and apply common distributions using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of common distributions into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies common distributions, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Common Distributions in one precise sentence and name one assumption.
2. Create a two-step example of common distributions and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 05 / CONCEPT

Expectation and Variance: Concept

01 OBJECTIVE

Explain and apply expectation and variance using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Expectation and Variance is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Expectation, Variance are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should expectation and variance be used, and what observable evidence would make that choice defensible? Build a small local example that applies expectation and variance, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Expectation and Variance in one precise sentence and name one assumption.
2. Create a two-step example of expectation and variance and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 05 / WORKFLOW

Expectation and Variance: Workflow

01 OBJECTIVE

Explain and apply expectation and variance using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Expectation and Variance is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to expectation and variance.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies expectation and variance, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Expectation and Variance in one precise sentence and name one assumption.
2. Create a two-step example of expectation and variance and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 05 / WORKED EXAMPLE

Expectation and Variance: Worked Example

01 OBJECTIVE

Explain and apply expectation and variance using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies expectation and variance, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns expectation and variance into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Expectation and Variance in one precise sentence and name one assumption.
2. Create a two-step example of expectation and variance and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 05 / FAILURE ANALYSIS

Expectation and Variance: Failure Analysis

01 OBJECTIVE

Explain and apply expectation and variance using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how expectation and variance can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to expectation and variance. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Expectation and Variance in one precise sentence and name one assumption.
2. Create a two-step example of expectation and variance and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 05 / APPLIED LAB

Expectation and Variance: Applied Lab

01 OBJECTIVE

Explain and apply expectation and variance using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of expectation and variance into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies expectation and variance, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Expectation and Variance in one precise sentence and name one assumption.
2. Create a two-step example of expectation and variance and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 06 / CONCEPT

Sampling: Concept

01 OBJECTIVE

Explain and apply sampling using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Sampling is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Sampling are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should sampling be used, and what observable evidence would make that choice defensible? Build a small local example that applies sampling, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Sampling in one precise sentence and name one assumption.
2. Create a two-step example of sampling and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 06 / WORKFLOW

Sampling: Workflow

01 OBJECTIVE

Explain and apply sampling using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Sampling is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to sampling.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies sampling, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Sampling in one precise sentence and name one assumption.
2. Create a two-step example of sampling and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 06 / WORKED EXAMPLE

Sampling: Worked Example

01 OBJECTIVE

Explain and apply sampling using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies sampling, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns sampling into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Sampling in one precise sentence and name one assumption.
2. Create a two-step example of sampling and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 06 / FAILURE ANALYSIS

Sampling: Failure Analysis

01 OBJECTIVE

Explain and apply sampling using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how sampling can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to sampling. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Sampling in one precise sentence and name one assumption.
2. Create a two-step example of sampling and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 06 / APPLIED LAB

Sampling: Applied Lab

01 OBJECTIVE

Explain and apply sampling using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of sampling into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies sampling, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Sampling in one precise sentence and name one assumption.
2. Create a two-step example of sampling and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 07 / CONCEPT

Estimation: Concept

01 OBJECTIVE

Explain and apply estimation using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Estimation is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Estimation are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should estimation be used, and what observable evidence would make that choice defensible? Build a small local example that applies estimation, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Estimation in one precise sentence and name one assumption.
2. Create a two-step example of estimation and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 07 / WORKFLOW

Estimation: Workflow

01 OBJECTIVE

Explain and apply estimation using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Estimation is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to estimation.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies estimation, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Estimation in one precise sentence and name one assumption.
2. Create a two-step example of estimation and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 07 / WORKED EXAMPLE

Estimation: Worked Example

01 OBJECTIVE

Explain and apply estimation using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies estimation, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns estimation into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Estimation in one precise sentence and name one assumption.
2. Create a two-step example of estimation and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 07 / FAILURE ANALYSIS

Estimation: Failure Analysis

01 OBJECTIVE

Explain and apply estimation using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how estimation can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to estimation. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Estimation in one precise sentence and name one assumption.
2. Create a two-step example of estimation and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 07 / APPLIED LAB

Estimation: Applied Lab

01 OBJECTIVE

Explain and apply estimation using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of estimation into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies estimation, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Estimation in one precise sentence and name one assumption.
2. Create a two-step example of estimation and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 08 / CONCEPT

Hypothesis Testing: Concept

01 OBJECTIVE

Explain and apply hypothesis testing using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Hypothesis Testing is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Hypothesis, Testing are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should hypothesis testing be used, and what observable evidence would make that choice defensible? Build a small local example that applies hypothesis testing, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Hypothesis Testing in one precise sentence and name one assumption.
2. Create a two-step example of hypothesis testing and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 08 / WORKFLOW

Hypothesis Testing: Workflow

01 OBJECTIVE

Explain and apply hypothesis testing using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Hypothesis Testing is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to hypothesis testing.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies hypothesis testing, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Hypothesis Testing in one precise sentence and name one assumption.
2. Create a two-step example of hypothesis testing and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 08 / WORKED EXAMPLE

Hypothesis Testing: Worked Example

01 OBJECTIVE

Explain and apply hypothesis testing using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies hypothesis testing, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns hypothesis testing into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Hypothesis Testing in one precise sentence and name one assumption.
2. Create a two-step example of hypothesis testing and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 08 / FAILURE ANALYSIS

Hypothesis Testing: Failure Analysis

01 OBJECTIVE

Explain and apply hypothesis testing using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how hypothesis testing can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to hypothesis testing. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Hypothesis Testing in one precise sentence and name one assumption.
2. Create a two-step example of hypothesis testing and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 08 / APPLIED LAB

Hypothesis Testing: Applied Lab

01 OBJECTIVE

Explain and apply hypothesis testing using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of hypothesis testing into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies hypothesis testing, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $\hat{y} = w * x + b$
Loss: $L = (\hat{y} - y)^2$
Gradient: $dL/dw = 2 * (\hat{y} - y) * x$
Update: $w_{next} = w - learning_rate * dL/dw$

05 PRACTICE

1. Define Hypothesis Testing in one precise sentence and name one assumption.
2. Create a two-step example of hypothesis testing and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 09 / CONCEPT

Calibration: Concept

01 OBJECTIVE

Explain and apply calibration using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Calibration is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. Begin by asking what problem the idea solves, what information enters the process, what result should leave it, and what evidence would show that the result is useful. In Probability and Statistics, the terms Calibration are not vocabulary to memorize in isolation; they describe a system of relationships. A strong learner can translate the idea between plain language, a diagram, a small numerical or code example, and a statement of limitations. This page establishes that translation before adding detail.

03 REASONING

Central question: When should calibration be used, and what observable evidence would make that choice defensible? Build a small local example that applies calibration, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Calibration in one precise sentence and name one assumption.
2. Create a two-step example of calibration and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 09 / WORKFLOW

Calibration: Workflow

01 OBJECTIVE

Explain and apply calibration using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Calibration is a central part of probability and statistics because it changes how inputs, assumptions, implementation choices, and evidence connect. A first-principles view separates the mechanism into state, operation, constraint, and test. State describes what is currently known. The operation transforms that state. A constraint rules out invalid moves, and a test compares the output with an expectation. This structure prevents an implementation from becoming a collection of unexplained steps. It also makes debugging local: identify which state, operation, constraint, or test failed. Apply this model directly to calibration.

03 REASONING

Workflow: define the smallest valid input; predict the next state; perform one operation; test one invariant; then expand. Example: Build a small local example that applies calibration, records the result, and tests one failure condition.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Calibration in one precise sentence and name one assumption.
2. Create a two-step example of calibration and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 09 / WORKED EXAMPLE

Calibration: Worked Example

01 OBJECTIVE

Explain and apply calibration using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Consider this task: Build a small local example that applies calibration, records the result, and tests one failure condition. First state the expected output before calculating or coding. Next choose a minimal representation and execute one step at a time. After every step, compare the intermediate state with the rule that should still hold. Finally, test a boundary case that differs from the example in exactly one important way. This worked-example pattern turns calibration into a repeatable method rather than a memorized answer.

03 REASONING

Worked reasoning: 1) restate the task; 2) identify knowns and unknowns; 3) choose the smallest method; 4) calculate or trace; 5) verify independently; 6) explain what would make the result fail.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Calibration in one precise sentence and name one assumption.
2. Create a two-step example of calibration and verify the intermediate state.
3. Name one boundary case, one likely misconception, and one test that would expose it.
4. Connect this page to a prior concept and explain why the connection matters.



UNIT 09 / FAILURE ANALYSIS

Calibration: Failure Analysis

01 OBJECTIVE

Explain and apply calibration using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Failure analysis asks how calibration can produce a plausible but wrong result. Common causes include an incorrect assumption, a representation that loses information, an edge case omitted from tests, an invalid comparison, or evidence taken from the wrong stage of the process. Change one factor at a time, preserve the failing example, and write a regression test before repairing the implementation. A clear failure record is part of the learning artifact.

03 REASONING

Failure probe: construct the smallest input that breaks the naive approach to calibration. State the expected behavior and why the naive result is misleading.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Calibration in one precise sentence and name one assumption.
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UNIT 09 / APPLIED LAB

Calibration: Applied Lab

01 OBJECTIVE

Explain and apply calibration using explicit inputs, assumptions, steps, evidence, and failure checks.

02 CORE EXPLANATION

Implementation converts the idea of calibration into inspectable behavior. Start with a deterministic example small enough to check by hand. Keep data preparation, core logic, and reporting separate. Add assertions for shape, type, range, or state as appropriate. Record the configuration and avoid relying on hidden global state. The goal is not merely to obtain output; it is to create an output whose origin can be explained and reproduced.

03 REASONING

Lab brief: Build a small local example that applies calibration, records the result, and tests one failure condition. Save the input, expected output, observed output, and a short explanation of any difference.

04 IMPLEMENTATION

Model: $y_{\text{hat}} = w * x + b$
Loss: $L = (y_{\text{hat}} - y)^2$
Gradient: $dL/dw = 2 * (y_{\text{hat}} - y) * x$
Update: $w_{\text{next}} = w - \text{learning_rate} * dL/dw$

05 PRACTICE

1. Define Calibration in one precise sentence and name one assumption.
2. Create a two-step example of calibration and verify the intermediate state.
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FILE / ORIENTATION

Deep-Dive Capstone

01 FILE GUIDANCE

Build a compact, reproducible artifact demonstrating probability and statistics. Include assumptions, a baseline, tests, failure analysis, results, limitations, and instructions for repeating the work offline.

02 LOCAL USE

Index this page in the offline database and preserve its resource and page identifiers for bookmarks, notes, highlights, and progress.